Digital Humanities projects by university students for pupils. Initial results and applicable tools.

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Introduction

Critical digital competences and digital skills are becoming increasingly important in lifelong learning (European Commission 2022). So far, digital humanities educational approaches have been primarily aimed at university students (Muenster et al. 2021). This paper explores digital education for pupils in a humanities teaching-learning lab. The aims of this article are, based on a mapping into European frameworks, (a) to present initial results on the participation of pupils in digital humanities projects by university students and (b) to reflect on the possibilities and limitations of using digital tools in humanities education programmes for pupils.

European framework programmes for education in the digital humanities

The European DigComp Framework (European Commission 2022) discusses digital competences as key competences for citizens in everyday life. Frameworks that contain recommendations for educators include the European Framework for the Digital Competence of Educators (DigCompEdu) (European Commission 2017) and the global Digital Literacy Framework (UNESCO 2018), of which the latter is aimed at university teachers in English-speaking countries and covers media literacy, communication and collaboration, study skills and information literacy, among others.

The involvement of pupils in digital humanities projects by students

In order to test the possibilities and limits of involving pupils in digital humanities projects, a teaching-learning lab (DHLab) was set up at Friedrich Schiller University Jena in 2022, where students are trained as tutors and design, run and evaluate courses for pupils. So far, 10 courses have been run in 6 local schools and leisure centres by 14 students from the fields of art history, film studies, history, politics, German studies, sociology, computer science and teaching. Approximately 110 pupils aged 7-16 have taken part in the courses, which mainly take place in extracurricular settings. Previous course topics have included digital source research on churches in Jena and “Stolpersteine”, 3D scanning of historical and contemporary everyday objects, creating virtual city tours and Instagram narratives on historical figures and digital eyewitness interviews on the future of Jena.

In 6 courses designed by students between May 2023 and July 2024, a total of 43 pupils aged 8-16 were each qualitatively surveyed twice on topics related to the DigComp Framework. The aim was to find out what previous experience the pupils already had and to what extent they had a basic awareness of online safety, for example. The data collected served as the basis for the student tutors' course planning.[1]

Initial results from the surveys cautiously suggest that teaching digital skills is also possible to some extent for younger age groups, albeit in a playful way and with the support of student tu-

tors. Awareness of internet safety was found in all age groups to varying degrees, but knowledge of specific safe websites and search engines was almost universally lacking. The ability to find relevant and reliable information on the Internet was also found to be rather low in all age groups, according to the individual assessment of the participants. However, it became comparatively higher with increasing age and experience.

The informative value of the survey is primarily limited by the small number of participants and the different questions, which varied in content, format and presentation depending on the age group and course topic.[2]

The use of digital tools in humanities courses for pupils

To prepare the students for their task of designing courses for pupils they were introduced to the contents of die DigComp Framework and provided with knowledge on suitable tools for imparting culture-historical topics digitally. This knowledge includes a for the DHLab especially curated tool list which focuses on digital tools and methods that can easily be implemented in teaching activities without expert knowledge. In contrast to other tool lists or data sets focussing on the context of cultural heritage (e. g. the Europeana tool list (Europeana 2024)) all tools are are described with examples of use in the context of teaching and easily accessible for varying groups of learners.

Name & link	Application possibilities
Jena4D / 4DCity-Browser https://4dcity.org Browser application Smartdevice application	- virtual historical townscapes and hyperlinks to further information and media - designing digitally enhanced city tours *with* pupils - stimulating participation through re-photography of historical photographs
3D Heritage https://3dheritage.de/ https://sketchfab.com/dfg3dviewer Browser application Smartdevice application	- what it is: collection of professionally and non-professionally digitized 3D-models - possibility to participate in citizen science and the digitization of cultural assets with the creation of new 3D-models via smartphone cameras - possibility to engage directly with the preservation of cultural assets

Figure 1. Applications and tools with local focus

Name & link	Application possibilities
Oral History Toolkit https://oralhistory.lib.uci.edu/	- toolkit and guidelines to create „oral history"-projects - introduction to methods of historical science and possibility to participate in citizen science - fitting for older pupils and learners - fitting for a project setting
TimeMapper http://timemapper.okfnlabs.org/	- tool for the creation of interactive maps with timelines and linkups to further information - possibility to learn about the connection between place and time with focus on historical events
Soundcite JS http://soundcite.knightlab.com/	- for incorporating audio tracks/oral history into text that is presented digitally on a website - Wordpress plugin available - possibility to connect audio narration and digital storytelling - suitable as addition to the *Oral History Toolkit*
Timeline JS http://timeline.knightlab.com/#make	- tool for creating timelines for the web-based presentation - low-threshold usability via Excel tables - possibility to learn about the connection between place and time with focus on historical events
Old Maps Online https://www.oldmapsonline.org/ Browser application	- vast collection of global historical maps - integrated tool to compare and geo-reference historical and contemporary maps - collections can be managed for an explorative approach

Figure 2. General applications for creating your own digital content

Over the course of the project, it could be observed that the student tutors favoured two tools and methods for teaching. (1) *4DCity-Browser* to create virtual city tours with their pupils and learn about the city's history (Muenster et al. 2024) (2) *3D Heritage / Sketchfab* to digitise three-dimensional objects to preserve and contextualise objects (Muenster et al. 2022).

In general students as well, pupils preferred approaches that left room for creative input and high degrees of personal participation which correlated with the findings of other educational sciences (Autenrieth / Nickel 2020). For our own teaching -learning lab we found is advisable to rely heavily on an explorative approach especially because most courses for pupils are conducted on an extracurricular base and are visited voluntarily by the learners.

On the one hand, this has the advantage that learning does not take place in a strict learning setting, but on the other hand it also means that learners can be introduced to scientific work and familiarise themselves with ways of participating independently in the preservation of cultural assets.